MIE 1624 Introduction to Data Science, Analytics and AI – Winter 2026

Course Project

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Deadline: Sunday, March 29, 11:59pm

Background

Global political crises, like russia's war against Ukraine and Trump's tariff threats, underscore the necessity for global innovations to accelerate economic growth, enhance competitiveness, and empower nations to resist aggressions. We need to review the state of the Canadian innovation ecosystem and develop a new strategy to facilitate innovations in Canada. The Artificial Intelligence ecosystem is an integral part of Canada's innovation strategy. In 2017, Innovation, Science and Economic Development Canada (ISED) launched the first phase of Pan-Canadian Artificial Intelligence Strategy under the leadership of CIFAR (Canadian Institute for Advanced Research), with the focus on Canada's talent base and global competitiveness in AI research. Multiple reports on the impact assessment of the Pan-Canadian AI Strategy have been published, including the 2020 report by Accenture and the 2023 report by CIFAR. Those reports did not take into account new challenges (COVID-19 pandemic, war in Ukraine, and policies of Trump's administration revealed in 2025) and new opportunities (development of Generative AI models such as ChatGPT) that appeared since the reports' publication.

On September 26th, 2025 the Government of Canada released a statement outlining the launch of a new AI strategy task force and a 30-day national sprint (in the month of October 2025) with the goal of generating ideas about Canada's AI strategy (Government of Canada launches AI Strategy Task Force and public engagement on the development of the next AI strategy - Canada.ca). The government is seeking advice on a range of AI-related themes, including: research and talent, AI adoption across industry and governments, commercialization of AI, scaling Canadian AI champions and attracting investments, building safe AI systems and strengthening public trust in AI, education and skills, building enabling infrastructure, security of the Canadian infrastructure and capacity.

Your team has been hired *pro bono* by the University of Toronto as consultants to evaluate Canada's position among other countries in terms of AI innovations, rewrite or enhance the current AI strategy draft, and propose practical steps to implement the proposed strategy. Your task is to analyze various traditional media, social media, reports, YouTube videos, and other sources of information in different countries to determine how Canada can accelerate AI adoption and improve AI competitiveness. You need to write a consulting report and deliver a presentation presenting your findings and your suggestions to the Government of Canada for changes in its current portfolio of AI strategies that are projected to have a positive impact on Canada's economic development and help Canada to secure peace and prosperity among other nations in the future.

Data science, analytics, AI, and big data are becoming increasingly prevalent in various fields, resulting in a growing demand for data analysts, data scientists, analytics and AI managers, and other data professionals. As a result, text analytics and natural language processing are now a hot topic among researchers, educators, and entrepreneurs. In addition, the design of questionnaires and surveys, combined with statistical analysis, trend building, and projections, allows us to perform advanced data analytics studies. The development of Generative AI models, such as ChatGPT, gives an opportunity to summarize text information and search for insights and ideas in large text datasets.

In this project, you need to perform an analysis of key factors responsible for differences in AI-related innovations across countries. To achieve this, your team will first need to collect public data from news sources, social media, reports, blogs, and interviews with experts on the development of AI ecosystems in various countries and their corresponding AI strategies. You need to summarize how AI innovations are measured in different countries and how countries are rated relative to each other by creating a quantitative AI score/ranking for each country. You can use the 2025 AI Index Report, published annually by Stanford University at https://hai.stanford.edu/ai-index/2025-ai-index-report, as a primary source for such a comparison. In addition, Tortoise Media publishes AI index/ranking of different nations based on their level of investment, innovation, and implementation of AI https://www.tortoisemedia.com/data/global-ai. For instance, one possible measure of AI innovations can be the number of unicorns (AI-focused startups valued at $1B or more) in each country. You need to analyze which indicators/factors (e.g., human capital, productivity, R&D spendings, AI research tax breaks, etc.) drive AI competitiveness, how Canada stands on those factors, and which factors can be critical to improving Canada's position relative to other countries.

Second, your team will need to develop an enhanced proposal for Canada's AI ecosystem development strategy. To do this, you will be using generative AI models to summarize experiences and extract key factors contributing to AI competitiveness of other countries from reports, strategy papers, scientific publications, social media (especially Twitter/X), expert opinions. Adjusting those strategies for a Canadian situation is also important. You can use reports published by ISED and CIFAR as a starting point for developing your strategy, shorten and enhance them. Recommended existing Canadian frameworks to analyze and propose improvements to include the $2 billion Canadian Sovereign AI Compute Strategy (2024) and the AI Strategy for the Federal Public Service 2025-2027. Alternatively, you may develop your AI strategy from scratch. Materials collected by the course instructor are available for you at the Google Drive folder https://drive.google.com/drive/folders/1jhev3A80L4INSqeyLM53ZhnKH2pnSktf.

Third, based on the results of your analysis, your team will need to develop data-driven practical steps (government programs, internship programs, industry-academic collaboration initiatives, international collaborations and collaborations with Canadians abroad, partnerships with multinational companies, tax incentives for R&D projects, defense funding guidelines, funding from international organizations and agencies, etc.) to implement the proposed AI strategy in

practice. All recommended changes to Canada's AI strategy need to be clearly contrasted against the current status quo, e.g., if you propose an increase to the budget, you must explain what the current budget is and what revenue sources would fund the budget increase. Your practical steps may be tied up with a PR strategy to promote Canada in media and social media as "Canada – the country of AI innovations". You may want to develop visualizations (e.g., infographics), key messages, and propose news stories to perform storytelling. The book "Ingenious: How Canadian Innovators Made the World Smarter, Smaller, Kinder, Safer, Healthier, Wealthier, and Happier" may give you some ideas related to storytelling for promoting Canadian AI innovations.

Finally, based on your analytics and results, you will create a chatbot that allows a user to have a meaningful conversation about Canada's AI strategy. You can use any Generative AI chatbot platform, e.g., LangChain based platforms such as CrewAI (Python module that you can install with `!pip install crewai`). You can explore courses on DataCamp and on other online course portals, such as Introduction to AI Agents and Building AI Agents with CrewAI courses. You are required to use Python to interface with the chatbot platform back-end and front-end user interface, LLM API (GPT-5.2, Gemini 3.0, Claude 4.5, Llama 3.1-4) and with your Python modeling code. Feel free to deploy your applications on Google Cloud, Microsoft Azure Cloud or Amazon AWS Cloud, if you decide to do so. The front-end to your chatbot can be Python (e.g., Streamlit module), any messenger (Facebook Messenger, WhatsApp, Slack, etc.), a web-site chat-window (Wordpress plug-in, etc.) or any other user interface.

Learning Objectives

- Develop the ability to work in a team on a consulting project. (You are required to work on the project in the same group as for your in-class presentation. Check the Quercus portal for the list of your group members.)
- Improve skills and competencies required for performing a full cycle of data science and analytics workflow, i.e., data collection and pre-processing, applying algorithms to analyze data, trend identification, storytelling based on analytics (writing a consulting report and delivering an oral presentation). This is an open-ended problem, and as such, creative solutions are encouraged where possible.

Tools Allowed

- You can use Python libraries mentioned in class as well as any other Python libraries you find during your research. Note that you can only use Python 3.
- If you plan to use other tools beyond Python, including but not limited to software products, cloud services, chatbots, during your research, save outputs of those to file(s) and read them into your Python code. You should not be restricted by Python limitations in this project.
- You can use any Generative AI-based chatbot platform interfaced from Python.
- For visualizing results in your report and presentation, you may use Python or any other outside tool, e.g., Excel, Tableau, Power BI, etc.

To Do Summary

Perform analytics on news articles, reports, social media posts, books, YouTube videos, search results, etc., to score/rank countries in terms of AI innovations and compare/contrast countries' policies with high scores against Canada to find out how Canada can innovate better in AI. You are not restricted in the sources of information that you can analyze, e.g., New York Times, Thomson Reuters, Associated Press, Facebook, Twitter/X, Instagram images, Amazon, Google search, and in the way you get access to that information, e.g., APIs or web-scraping. (You may want to perform automatic or manual fact-checking of the information that you analyze to assess your data's reliability.) Use of Generative AI tools, such as ChatGPT, is encouraged. You may feed analyst reports and other relevant documents as PDF/DOCX files to Generative AI models (GPT, Gemini, Claude, etc.) to summarize documents and to extract ideas for your strategy document. You may classify or cluster information about different topics related to AI innovations in Canada and other countries, and then try to cover all the identified topics or concentrate on only some of them. You can use the algorithms covered in the course as well as any other algorithms that you find during your research. (Feel free to use cloud services on Amazon AWS, Google Cloud, Microsoft Azure, IBM Cloud for your analysis or data retrieval.)

Among others, you may consider utilizing the following algorithms for this project:

- Sentiment analysis of social media posts and news articles related to AI innovations.
- Factor and topics identification, e.g., based on your sentiment analysis results, you may try to identify factors/reasons/topics that drive sentiment.
- Clustering, e.g., clustering of practical steps and governmental programs that support AI innovations in different countries.
- Optimization models, e.g., optimizing government funding and budgets to support AI innovations.
- Trend building with regression models.
- Classification algorithms.
- Generative AI algorithms.

Note: the scope of the project is quite wide, and it is advised that you narrow it down based on your interests and expertise. Make the project truly yours.

TO DO

Finish the following five parts **based on your data science and AI-based analytics:**

Part 1 – Collect data and summarize how AI innovations and AI competitiveness are measured:

Collect public data from news, social media, reports, blogs, and interviews with experts about the development of AI ecosystems in different countries and their AI strategies. You need to summarize and extract key information on how AI innovations and AI competitiveness are measured in different countries. Use this information to generate an AI innovation/competitiveness score/ranking for each country and use this to compare/contrast policy strategies that correlate with

better AI competitiveness. You can use 2025 AI Index Report, published annually by Stanford University at https://hai.stanford.edu/ai-index/2025-ai-index-report as well as Tortoise Media Global AI Index, as primary sources of such comparison. For instance, one possible measure of AI innovations can be the number of unicorns (AI-focused startups valued at $1B or more) in each country. You need to analyze which indicators/factors (e.g., human capital, productivity, R&D spendings, AI research tax breaks, etc.) drive AI competitiveness, how Canada stands on those factors, and which factors can be critical to improving Canada's position relative to other countries.

Another recent publication that you may find interesting is "Study shows Canada among least AI literate nations" produced by KPMG in 2025. The study found Canada ranked 44th in AI training and literacy out of 47 countries. In September 2024, former European Central Bank president Mario Draghi presented a report The Future of Europe's Competitiveness, which was commissioned by the European Commission. His central proposal is to mobilize investments of up to €800 billion per year to regain ground lost to the United States and China in terms of economic growth, innovation, and productivity, including in the field of AI.

If you find that the dataset provided by the course instructor in the Google Drive folder at https://drive.google.com/drive/folders/1jhev3A80L4INSqeyLM53ZhnKH2pnSktf is insufficient for your purposes, feel free to use additional datasets available on the web (send new documents to the course instructor or TAs if you would like those to be added to the Google Drive folder). Among other datasets, you may consider:

- *Opinion of influencers* – tweets of people that discuss AI innovations (you may use Twitter/X API from Python to collect tweets related to AI innovations);
- *Social media analysis* (sentiment, topic, AI innovation, and AI competitiveness measures);
- *Kaggle datasets collected by organizations or people*, e.g., https://www.kaggle.com/datasets/muhammadroshaanriaz/ai-and-ml-popularity-indexanalyzing-global-trends, https://www.kaggle.com/datasets/katerynameleshenko/ai-index and https://www.kaggle.com/datasets/paultimothymooney/ai-index-report-2025;
- *YouTube videos*, e.g., http://youtu.be/LdZoM3n4TJM, https://youtu.be/slqS4UUSQYo and http://youtu.be/PNtSmpGtsr0;
- *Podcasts*, e.g., https://www.cbc.ca/radio/frontburner/canadas-bet-on-an-ai-boom-9.6937185
- *Data World and other "open data" datasets* https://data.world/search?q=AI;
- *News aggregators* (API or web-scraping), e.g. https://www.pressreader.com/search?query=AI&orderBy=Relevance&searchFor=Articles;
- *Google Trends* https://trends.google.com, Kaggle, and similar data/news aggregation portals.

Feel free to experiment with Python libraries and APIs to download news articles, e.g., https://newsapi.org/docs/client-libraries/python, and Python libraries such as `PyPDF2` and `PyMuPDF` to convert PDF files to text.

Summarize results from your datasets, e.g., with Generative AI models (GPT, Gemini, Claude) using APIs from Python, or user interfaces such as https://chat.openai.com. For each dataset, make conclusions about your results. Explain those. Select summaries of datasets that you plan to use in Part 2.

Part 2 – Develop proposal for Canada's AI Strategy:

Your team will need to come up with an enhanced proposal about Canada's AI strategy. You can use reports published by ISED and CIFAR as a starting point for developing your strategy, shorten and enhance them. Your analysis should address how Canada's existing frameworks (Canadian Sovereign AI Compute Strategy (2024) and the AI Strategy for the Federal Public Service 2025-2027) could be improved or expanded. Alternatively, you may develop your strategy from scratch. You are not restricted in how you structure your report and presentation, but possible sections of the AI strategy proposal should include results of Part 1, Part 2, Part 3, and Part 4. Please include Part 5 as an appendix in your report and as a demo video in your presentation.

Summarizing experiences of other countries, which you performed in Part 1 with generative AI models from reports, strategy papers, scientific publications, social media (especially Twitter/X), expert opinions, may be a key point in your analysis. Adjusting those strategies for a Canadian situation is also important.

Summarize the results of your analysis in the form of a consulting report and presentation slides.

Part 3 – Practical steps for implementing your AI strategy:

Based on your analysis, your team will need to develop practical steps (government programs, internship programs, industry-academic collaboration initiatives, international collaborations and collaborations with Canadians abroad, partnerships with multinational companies, tax incentives for R&D projects, defense funding guidelines, funding from international organizations and agencies, etc.) to implement your proposed AI strategy in practice. As an example of a practical step, you may analyze Canada's Mitacs research internship program and propose enhancements or modifications to it. Your strategy and practical steps may be tied up with a PR strategy to promote Canada in media and social media as "Canada – the country of AI innovations", that you perform in Part 4. Each recommendation must include: estimated cost, implementation timeline (1-3 years), responsible ministry or agency, and one specific metric for measuring success.

Part 4 – Visualizations, storytelling, recommendations:

Your task is to visualize modeling results obtained in Part 1, 2 and 3. Design visualization(s), e.g., wordclouds, that allows decision makers to grasp the current state of AI ecosystem in Canada, key findings and milestones of your proposal, practical implementation steps that you propose, etc. Your strategy and practical steps should be tied up with a PR strategy to promote Canada in media and social media as "Canada – the country of AI innovations". You may want to develop visualizations (e.g., infographics), key messages, and propose news stories to perform storytelling based on your analysis.

In addition, based on your analysis and identification of key factors/reasons/topics/steps you need to develop a narrative (storytelling via presentation and report) presenting your findings and your suggestions to the Government of Canada, and national and international NGOs for applying for funding to support the portfolio of strategies that your team developed. Your storytelling should concentrate on having a projected positive impact on Canada's international presence and image. Your projection should include how Canada's AI innovation/competitiveness score will improve over time as a result of implementing your policy changes. If necessary, you may complement your analysis with posts/tweets templates that you propose to publicize your strategy. Your presentation must focus on exactly 3 key recommendations (which may be multi-faceted). Additional recommendations may be included in your written report.

Part 5 – LLM chatbot:

Based on your modeling results obtained in Part 1, 2 and 3, and your storytelling results and recommendations from Part 4, develop a chatbot that allows a user to have a meaningful conversation about Canada's AI strategy. You can use any Generative AI chatbot platform, e.g., LangChain based platforms such as CrewAI (Python module that you can install with `!pip install crewai`). Your results from Parts 1-4 need to be provided as a context to an LLM model, either directly or as embeddings retrieved from a vector database (LangChain, RAG). Your chatbot implementation must demonstrate at least TWO of the following capabilities: **(1)** multi-agent collaboration (e.g., separate researcher, analyst, and writer agents), **(2)** tool use beyond document retrieval (e.g., web search, code execution), or **(3)** agentic workflows with planning, iteration, or self-correction. Simple RAG-only implementations will receive a maximum of 3 out of 6 points for the Analysis component.

Note: the scope of the questions in Part 1-5 is quite wide, and it is advised that you narrow it down based on your interests and expertise. Make the work truly yours.

Project Presentations

- Project presentations are scheduled for **Tuesday, March 31, 6:00-9:00pm (room BA 1130)**, doors are open to the public.
- **Do not make your presentation overly technical**. Your audience is business-oriented and may know little about data science, people are interested in the insights that you got from your analysis and why your results can and should be used for decision-making.

What to Submit via Quercus

1. Jupyter notebook with appropriate documentation for every step as well as the relevant data files. Comment out any data retrieval processes (e.g., from web scraping, downloading, APIs, etc.) in your code and replace it with code for reading the corresponding data from files. If you decide to do some of your analysis outside of Python, save your results to data file(s) and read those into your Jupyter notebook. (**Submit all**

those data files together with your Jupyter notebook). Consider the Jupyter notebook as what you would report to senior data scientists and machine learning engineers. Documentation and comments in the Jupyter notebook should contain technical details of your data analysis and be understandable by data science professionals if they run it on their own. Make sure that your Jupyter notebook runs on Google Colab https://colab.research.google.com portal and that all needed data files are included in your submission. If the size of the data files exceeds Quercus's capacity, those should be stored on a cloud drive (e.g., Dropbox, Google Drive), and the link to the directory should be included in the notebook. If you use a LangChain framework such as Langflow or n8n for your LLM chatbot implementation, please export and include JSON file with the chain setup.

2. A 5 to 10-page consulting report in PDF and DOCX formats that summarizes your findings and results (all graphs should have axes appropriately labelled, all visual materials should be understandable, and the graphics should be of sufficient quality to be easily readable). This report should be business-oriented and cover your problem more extensively than your presentation.
3. Your business-oriented presentation slides in PowerPoint and PDF formats. Each group will deliver a 7-minute presentation followed by 1-2 minutes of Q&A handled by your AI chatbot. Presentations will be timed and stopped after 7 minutes. Use this structure: Problem (1 min), Key Findings (1-2 min), 3 Recommendations (3 min), Call to Action (1-2 min). It is up to you how many group members are presenting (one, two or all).
4. Video recording of your ~7-8 minute presentation in mp4 or avi format. You do not need a video recording of you from a webcam, audio recording on-top of your slideshow presentation (screenshare) is fine. The video recording should include 1-2 minutes of Q&A with your AI chatbot, based on your own set of questions.

Notes

- For the deliverables, consider the Jupyter notebook as what you would report to senior data scientists and machine learning engineers, and the consulting report and the presentation as what you would report to PR managers, university officials, government officials, journalists, and general audience.
- The presentation would be a visual representation of the executive summary of your report.
- The audience for your presentation and report in particular is business-oriented and includes people who are interested in the insights you gathered from your analysis and how your results should be used for decision-making.
- If embedding videos in your presentation slides (e.g., chatbot demos), ensure the video file is in MP4 format and embedded directly in PowerPoint rather than linked. Test playback on several computers beforehand, and have a backup plan (e.g., YouTube link) in case of technical issues. You will be allowed to use your own computer for delivering the presentation.

Marking

- The **project is worth 20 points** (10 points for your business-oriented presentation and 10 points for your analysis and report).
- The **presentation** will be graded as follows (**10 total pts**):
  1. *Organization & Delivery* (**3 pts**)
     - Clear structure and logical flow, with an introduction, main content, and conclusion;
     - Demonstrated enthusiasm and poise throughout the presentation;
     - Effective time management ensuring all topics are covered without rushing within the given time frame.
  2. *Content* (**3 pts**)
     - Use of appropriate, impactful visuals that complement and enhance the spoken content;
     - Clear representation of high-level ideas, ensuring understanding for all audience levels;
     - Proactive and comprehensive answering of questions, showcasing depth of understanding.
     - Your AI chatbot will answer 2-3 questions from the audience about your methodology and recommendations, demonstrating depth of understanding.
  3. *Business Pitch* (**4 pts**)
     - Concrete recommendations backed by data and analysis;
     - Clear problem-solution correspondence, showing how the solution addresses the identified issue;
     - Tailored relevance to the target audience, ensuring the pitch resonates and is compelling.
- The **analysis** in **Jupyter notebook** and the **report** will be graded as follows (**10 total pts**):
  1. *Problem Definition & Data Search* (**1 pt**)
     - Skillful narrowing down of the broader problem to a focused research question;
     - Clearly defined problem statement that is concise yet comprehensive;
     - Efficient identification and sourcing of relevant data, ensuring data integrity.
  2. *Analysis* (**6 pts**)
     - Cleaning the data;
     - Relevant and insightful visualizations that aid understanding;
     - Proper application of algorithms, evaluation of algorithms' accuracy, and showcasing technical proficiency.
     - LLM chatbot implementation.
  3. *Discussion & Insights* (**3 pts**)
     - Clear and direct link between the analysis conducted and the problem identified;
     - Utilization of effective data-driven decision-making criteria, demonstrating critical thinking;
     - Engaging storytelling that makes the data and analysis relatable and memorable.
- Every group member gets the same mark for the project. It is your responsibility to determine how you split the work inside your group. At least half of your group needs to be present during the in-class project presentations to answer questions.